Shota Kakiuchi

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EDUCATION

Columbia University — New York, NY
Master of Science in Mechanical Engineering — May 2027

The George Washington University — Washington, D.C.
Bachelor of Science in Mechanical Engineering, GPA: 3.5 / 4.0 — May 2026
Concentration in Aerospace and Robotics

TECHNICAL SKILLS

Programming: Python, MATLAB, C++, SQL, Linux, Docker, ROS, Jupyter Notebook, State-Space Control, Kalman Filtering
Engineering Tools: SolidWorks (CAD), ANSYS (FEA, CFD), Jupyter Notebook, Raspberry Pi, Microcontrollers, Grafana
Technical Expertise: Manufacturing & Process Optimization, Optics & Metrology, DFM, Data Analysis & Visualization, Robotics
Languages (Native): English, Japanese

WORK EXPERIENCE

Apple — Japan
Hardware Engineering Intern – Optics & Camera — Jun 2026 - Aug 2026

- Evaluated and supported implementation of **new precision optical metrology technology**, developing measurement workflows and calibration procedures for 3D surface characterization of optical components.
- Conducted **sensitivity studies and controlled experiments**, applying MATLAB and 3D surface data analysis to identify systematic measurement errors, investigate root causes, and validate improvements in accuracy and repeatability. Collaborated with **hardware engineers, manufacturing teams, and suppliers** to communicate technical findings, document testing procedures, and support integration of new optical measurement capabilities.

The George Washington University — Washington, D.C
Undergraduate Research Assistant — Apr 2023 - May 2026

- Conducted **experimental fluid dynamics and rheometry** research, characterizing heartbeat-like pulsatile flow and fluid behavior for potential **biomedical device applications**.
- Applied **computational fluid dynamics (CFD), statistical methods, signal processing, and Python** to analyze experimental datasets, characterize fluid behavior, and visualize research findings.

Deloitte Consulting LLP — New York, NY
Cybersecurity Consulting Intern — Jun 2025 - Jul 2025

- Supported **Oracle ERP implementation and enterprise risk management**, evaluating security configurations, internal controls, and operational processes to identify risks and recommend improvements.
- Collaborated with cross-functional U.S. and Japanese client teams to translate technical requirements into **process optimization recommendations**, supporting project execution and stakeholder coordination.

Medtronic — Fort Worth, TX
Manufacturing Intern — Jun 2024 - Aug 2024

- Developed and implemented **RFID-based inventory tracking technology** to improve manufacturing traceability, optimize inventory workflows, and address approximately **$85K in annual inventory losses**.
- Applied **Lean Six Sigma and process optimization** to implement manufacturing workflow improvements, resolve production bottlenecks, and recover backlogged orders, supporting an estimated **$120K in annual labor savings.**

PROJECTS AND RESEARCH

Capstone - Rocket Airbrake System

- Led **flight computer architecture, avionics integration, and hardware prototyping**, integrating microcontrollers, sensor interfaces, power circuitry, and servo-actuated airbrakes for an altitude-targeting rocket.
- Developed **state-space estimation, Kalman filtering, sensor fusion, and active control algorithms**, leveraging MATLAB/Simulink modeling and flight data analysis to evaluate and validate apogee reduction.

IoT Research Project
Undergraduate Research Fellow

- Selected as the **sole engineering student among 12 fellows** to develop a patent-pending, **low-cost embedded IoT prototype**, integrating Raspberry Pi, sensors, and hardware components for scalable data acquisition.
- Developed an **automated, containerized data pipeline** using Docker, MQTT, Node-RED, InfluxDB, and Grafana to support real-time sensor monitoring, data processing, and visualization.